

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCIES302 - MECHANICS OF SOLIDS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 For most metals, Poisson's ratio lies between
A) 0.0 – 0.1 B) 0.1 – 0.3 C) 0.25 – 0.35 D) 0.4 – 0.6
- 2 The relation between E, G (shear modulus) and ν is
A) $E=2G(1+\nu)$ B) $E=G(1+\nu)$ C) $E=3G(1-2\nu)$ D)
 $E=9kG/(3k+G)$
- 3 BMD of a simply supported beam with a central point load is
A) Parabolic with max at center B) Triangular
C) Uniform D) Zero everywhere
- 4 In BMD, the point of contra-flexure is where
A) Shear force is zero B) Bending moment is zero but changes sign
C) Load is maximum D) Deflection is zero
- 5 The value of maximum bending moment in a simply supported beam with a central point load is
A) $WL/2$ B) $WL/4$ C) $WL^2/8$ D) $WL^2/4$.
- 6 The maximum shear force in a cantilever beam loaded with a uniformly distributed load occurs at
A) Free end B) Mid-span C) Fixed end D) Point of zero load
- 7 The relationship between shear force V and bending moment M at a section is
A) $V = dM/dx$ B) $M = dV/dx$ C) $V = Mx$ D) $V = M/x$
- 8 A point of contra-flexure is where
A) Shear force is zero B) Shear force is maximum
C) Bending moment changes sign D) Loading is maximum
- 9 The ratio of maximum shear stress to average shear stress is $4/3$ in -----
A) Circular cross-section B) Rectangular cross-section
C) Square cross-section D) All of the above

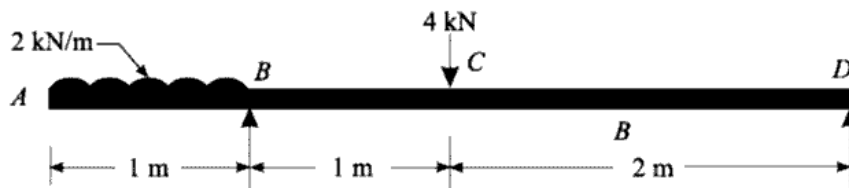
- 10 The strength of a beam depends upon
 A) Its section modulus B) bending stress C) Both (A) and (B) D) None of these
- 11 For two shafts joined in parallel, the ----- in each shaft is the same.
 A) shear stress. B) Angle of twist C) torque D) torsional stress.
- 12 In a simple bending theory, one of the assumptions is that the material of the beam is isotropic. This assumption means that the
 A) Normal stress remains constant in all directions
 B) Normal stress varies linearly in the material
 C) Elastic constants are same in all the directions
 D) Elastic constants varies linearly in the material

Q2. Solve the following.

- A) Derive the relationship between modulus of elasticity and modulus of rigidity. 6
- B) A copper rod AB is fixed at both ends of length 600 mm. When the temperature of the rod is 25°C , the gap of BC is 0.6 mm Determine
 1. Thermal stresses
 2. Temperature at which the gap will just close
 3. Stress and strain in the rod when its temperature is 100°C
 Take $E=100 \times 10^3 \text{ N/mm}^2$ and $\alpha=16 \times 10^{-6}/^{\circ}\text{C}$.

Q3. Solve the following.

- A) A beam 4 m long is overhanging by 1 m and carries a load as shown in the figure. 6
 Calculate the shear force and bending moment at important locations and draw the SFD and BMD for the beam. Also, locate the point of contraflexure.



- B) A simply supported beam AB is 10 m long. It carries a uniformly distributed load of 20 kN/m over a distance of 5 m from the left end A, a clockwise moment of 50 kN-m at 5 m, and a point load of 40 kN at a distance of 8 m from the left end A. Find shear force & bending moment at important locations and draw S.F.D. and B.M.D. for the beam.

Q4. Solve Any Two of the following.

- A) What is pure bending? Derive the bending stress equation. 6
- B) A beam has a T-section with the flange as well as the web 150 mm x 50 mm. If it is used as a simply supported beam and the bending moment at a section is 3400 Nm. Find the stresses at the extreme fibers of the beam. 6

- C) An I-section with rectangular ends has the following dimensions: 6
Flanges 150 mm x 20 mm and Web 300 mm x 10 mm
Find the maximum shear stress developed in the beam for a shear force of 100 kN.
Also draw a shear stress distribution diagram.

Q5. Solve Any Two of the following.

- A) A masonry pillar square in section 600 mm x 600 mm is subjected to a point load of 1800 kN at an eccentricity of 200 mm along one of the centroidal axes of the cross section. Find the stresses at four corners. Also determine the maximum eccentricity if the permissible tensile stress in masonry is limited to 2 N/mm².
- B) The stresses at a point of a machine component are 150 MPa and 50 MPa, both tensile. Find the intensities of normal, shear, and resultant stresses on a plane inclined at an angle of 55° with the axis of major tensile stress. Also find the magnitude of the maximum shear stresses in the components.
- C) At a certain point, the values of normal stresses across two planes at right angles to each other are $\sigma_x = 100$ x MPa and $\sigma_y = 40$ y MPa, both tensile and a shear stress, $\tau = 50$ MPa. Analyse either analytically or Mohr's Circle method and find:
1. The principal stresses and their locations.
2. Maximum Shear Stresses and their planes

Q6. Solve Any Two of the following.

- A) What is polar modulus, and derive the torsional formula. 6
- B) Using Euler's equation for long columns, determine the critical stresses for a compression member of slenderness ratio 80, 120, 160, and 200. The compression member has the following end conditions: (i) both ends hinged and (ii) one end hinged and the other end fixed. $E = 2 \times 10^5$ N/mm².
- C) A 1.5 m long column has a circular cross section of 50 mm diameter; one of the ends of the column is fixed in direction and position, and the other is free. Taking the factor of safety as 3. Calculate the safe load using i) Rankine's Formula with $f_c = 560$ N/mm² and $a = 1/1600$ for the pinned end ii) Euler's Formula with E for column = 1.2×10^5 N/mm².

***** End *****